

2025 Linear Algebra and Calculus Resit

① Compute $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$.

Since we're approaching $x=0$, $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + O(x^5)$, so $\lim_{x \rightarrow 0} \frac{x - \frac{x^3}{6} + \frac{x^5}{120} + O(x^5) - x}{x^3} = \lim_{x \rightarrow 0} \frac{-\frac{x^3}{6} + \frac{x^5}{120} + O(x^5)}{x^3}$

$$= \lim_{x \rightarrow 0} -\frac{1}{6} + \frac{x^2}{120} + O(x^2) = \boxed{-\frac{1}{6}}$$

② Define $f(x) = \begin{cases} x^2 \sin(\frac{1}{x}), & x \neq 0 \\ 0, & x = 0 \end{cases}$

a) Show that f is continuous at $x=0$.

For f to be continuous, we need to see that $f(0) = \lim_{x \rightarrow 0} f(x)$. We have $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x^2 \sin(\frac{1}{x})$. Recall that

$-1 \leq \sin(\frac{1}{x}) \leq 1 \quad \forall x \in \mathbb{R}$, so $-x^2 \leq x^2 \sin(\frac{1}{x}) \leq x^2$, and since $\lim_{x \rightarrow 0} x^2 = \lim_{x \rightarrow 0} -x^2 = 0$, by the Squeeze Theorem, $\lim_{x \rightarrow 0} x^2 \sin(\frac{1}{x}) = 0 = f(0)$.

b) Is f differentiable at $x=0$?

For that, we need to see whether $f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 \sin(\frac{1}{h}) - 0}{h} = \lim_{h \rightarrow 0} h \sin(\frac{1}{h}) = 0$ again, by the Squeeze Theorem.

Since the limit exists, f is differentiable.

③ $f_1(x, y) = \frac{x^2 + y}{y+1}$, $f_2(x, y) = x \sin(xy)$. Compute the Jacobian of $F = (f_1, f_2)$.

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{2x}{y+1} & \frac{y+1-x^2-y}{(y+1)^2} \\ \sin(xy) + x \cos(xy)y & x^2 \cos(xy) \end{bmatrix} = \begin{bmatrix} \frac{2x}{y+1} & \frac{1-x^2}{(y+1)^2} \\ \sin(xy) + xy \cos(xy) & x^2 \cos(xy) \end{bmatrix}$$

④ $f(x) = e^{-x} - x$

a) Show that f has one root in $[0, 1]$.

$f(0) = e^0 - 0 = 1 > 0$ and $f(1) = e^{-1} - 1 = \frac{1}{e} - 1 < 0$. Since f is continuous for being a difference between an exponential and a linear function, by Bolzano's Theorem, there is a root in $(0, 1)$.

b) $\lim_{x \rightarrow \pm \infty} f(x)$

$$\lim_{x \rightarrow \infty} f(x) = e^{-\infty} - \infty = -\infty, \quad \lim_{x \rightarrow -\infty} f(x) = e^{\infty} - \infty = \infty$$

c) $f'(x) = -e^{-x} - 1$.

d) Can there be other roots? No, because $f'(x) < 0 \quad \forall x \in \mathbb{R}$, which means the function always goes down. Also $\lim_{x \rightarrow \infty} f(x) = -\infty$ and $\lim_{x \rightarrow -\infty} f(x) = \infty$, which reinforces that f doesn't go back to the x -axis at any point.

⑤ $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a linear map defined by $T(x, y, z, w) = \begin{bmatrix} x+y+z+w \\ y+z \\ x+z+w \end{bmatrix}$.

a) Give the standard matrix A of T . $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$.

b) Basis for $\text{Nul } A$.

For that, we need to find solutions to $Ax=0$; $\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$

$\rightarrow \begin{cases} x+w=0 \\ y=0 \\ z=0 \end{cases} \begin{cases} x=-w \\ y=0 \\ z=0 \end{cases} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = w \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$. So a basis is $\left\{ \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$.

c) Basis for $\text{Col } A$.

A basis for $\text{Col } A$ will be given by the pivot columns of A . Following the row reduction in b), we see the first three columns are pivot columns, thus, a basis for $\text{Col } A$ is $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$.

d) Rank and nullity of T .

$\text{rank } T = \dim \text{Col } A = 3$, $\text{nullity } T = \dim \text{Nul } A = 1$.

e) Is T injective? Surjective?

T would be injective if $Ax=0$ only had the trivial solution, that is, if $\text{Nul } A = \{0\}$, but it's not the case, so T is not injective. It is surjective because $\text{rank } T = 3$, which is the dimension of the codomain ($\mathbb{R}^3$), that is, the columns of A span $\mathbb{R}^3$. This can also be seen since A has a pivot position in every row.

⑥ $A, B \in M_n(\mathbb{R})$, A invertible.

a) Solve $AXA^T = B$, for X . Since A is invertible, $A^T A X A^T = A^T B \rightarrow X A^T A = A^T B A \rightarrow X = A^T B A$.

b) Show that if B is diagonalizable, so is X .

If B is diagonalizable, then there are matrices P invertible and D diagonal such that $B = PDP^T$. We must show the same for X . We know $X = A^T B A$. Substituting: $X = A^T (PDP^T) A = (A^T P) D (P^T A)$ by associativity of the product of matrices. Now using a property about inverses of products: $X = (A^T P) D (A^T P)^T$, taking $Q = A^T P$ we have $X = Q D Q^T$, Q invertible and D diagonal, so X is diagonalizable.

c) Show X and B have the same eigenvalues.

That's the case when $\det(X - \lambda I) = \det(B - \lambda I)$. We can see that $\det(X - \lambda I) = \det(A^T B A - \lambda A^T A) = \det(A^T (B - \lambda I) A) = \det A^T \cdot \det(B - \lambda I) \cdot \det A = \det A^T \det A \cdot \det(B - \lambda I) = \det(A^T A) \cdot \det(B - \lambda I) = \det I \cdot \det(B - \lambda I) = \det(B - \lambda I)$.

⑦ $V = \{ f \in C^1([0, 1]) \mid f(0) = f(1) = 0 \}$. Show V is a vector subspace of $C^1([0, 1])$.

For that, we need to show three things:

- $\vec{0} \in V$. The zero is the function $f(x) = 0 \forall x \in \mathbb{R}$. Clearly, $f(0) = f(1) = 0$.

- Let $f, g \in V$, does $f+g \in V$? For that, $(f+g)(0) = f(0) + g(0) = 0 + 0 = 0$ and $(f+g)(1) = f(1) + g(1) = 0 + 0 = 0$, so yes.

- Let $f \in V$ and $c \in \mathbb{R}$, does $cf \in V$? For that, $(cf)(0) = c \cdot f(0) = c \cdot 0 = 0$ and $(cf)(1) = c \cdot f(1) = c \cdot 0 = 0$, so yes.

8) $A = \begin{bmatrix} 1 & \alpha & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \alpha \in \mathbb{R}$.

a) Eigenvalues of A .

Since A is triangular, the eigenvalues can be read off the diagonal, $\lambda_{1,2} = 1, \lambda_3 = 2$. Alternatively, one must solve $\det(A - \lambda I) = 0$ for λ .

b) For which α is A diagonalizable?

For those in which the dimension of each eigenspace matches the multiplicity of the corresponding eigenvalue. For that we compute bases of the eigenspaces by solving $(A - \lambda I)x = 0$ for each λ .

$$A - I = \begin{bmatrix} 0 & \alpha & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{cases} \alpha x_2 = 0 \\ x_3 = 0 \end{cases} \text{ If } \alpha = 0, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{ space has dimension 2.}$$

$$\text{If } \alpha \neq 0, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \text{ space has dimension 1.}$$

$$A - 2I = \begin{bmatrix} -1 & \alpha & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -\alpha & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 \text{ free} \end{cases} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \text{ space has dimension 1.}$$

So A diagonalizes only if $\alpha = 0$.

9) $v_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

a) Compute an orthogonal basis.

Since the vectors are linearly independent, we can apply the Gram-Schmidt process:

$$u_1 = v_1$$

$$u_2 = v_2 - \text{proj}_{u_1}(v_2) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{3}{5} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 6/5 \\ 3/5 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/5 \\ 2/5 \\ 0 \end{bmatrix}$$

$$u_3 = v_3 - \text{proj}_{u_1}(v_3) - \text{proj}_{u_2}(v_3) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{v_3 \cdot u_1}{u_1 \cdot u_1} u_1 - \frac{v_3 \cdot u_2}{u_2 \cdot u_2} u_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{2}{5} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \frac{4/5}{30/25} \begin{bmatrix} -1/5 \\ 2/5 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 4/5 \\ 2/5 \\ 0 \end{bmatrix} - \frac{20}{30} \begin{bmatrix} -1/5 \\ 2/5 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 4/5 \\ 2/5 \\ 0 \end{bmatrix} - \begin{bmatrix} 4/3 \\ 4/3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/3 \\ -2/3 \\ 1/3 \end{bmatrix}. \text{ So the orthogonal basis is } \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1/5 \\ 2/5 \\ 0 \end{bmatrix}, \begin{bmatrix} 1/3 \\ -2/3 \\ 1/3 \end{bmatrix} \right\}.$$

b) Normalize it.

$$\|u_1\| = \sqrt{5}, \|u_2\| = \frac{\sqrt{30}}{5}, \|u_3\| = \frac{\sqrt{6}}{3}, \text{ so the orthonormal basis is } \left\{ \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix}, \begin{bmatrix} -1/\sqrt{30} \\ 2/\sqrt{30} \\ 0 \end{bmatrix}, \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} \right\}$$

10) True or false?

a) If $P_{C \leftarrow B}$ is a change of basis matrix, then $\det P_{C \leftarrow B} = 0$.

False. Since $P_{C \leftarrow B}$ is made out of the coordinates of the vectors in a basis, the columns of $P_{C \leftarrow B}$ are linearly independent, thus $\det P_{C \leftarrow B} \neq 0$.

b) $\dim P_n = n$.

False. $\dim P_n = n+1$, since any basis will contain terms with all powers $\leq n$, including the 0, which makes $n+1$ elements in a basis.

c) If $u \cdot v = 0$, then u or v is zero.

False. $u \cdot v = 0$ only means they are orthogonal. As an example, $u = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, here $u \cdot v = 0$.